Calibrating Jump and Stochastic-Volatility Models to Bitcoin Options, in the Browser

Model composition, Fourier pricing, loss design, optimisation and validation

Mohammed-Yassine Barnicha

October 2026

Abstract

This report documents an application that prices, calibrates and analyses Deribit Bitcoin options with 28 affine models, entirely in a web browser. Each model is the composition of a variance block (Black–Scholes, Heston, double Heston, Heston with variance jumps) and a jump law (none, Merton, bi-lognormal, Kou, CGMY, Variance Gamma, NIG); all are priced through their characteristic function by the COS and PROJ methods, re-implemented in JavaScript and checked against a Python reference to $6 \cdot 10^{-6}$ \$. We derive the characteristic functions and their martingale constraints, the two Fourier methods with their error control, the loss functions and the vega weighting used in production, the optimisers (classic and adaptive Nelder–Mead, differential evolution), the multi-day calibration, the model Greeks, the gamma exposure and the Monte Carlo engine for exotics. On 26 August 2026 the Bates model fits 222 out-of-the-money options over six maturities with an implied-volatility RMSE of 37.3 bp. The application is at `https://yassinebarnicha.github.io/calibration-btc-svjd/`.

Contents

1 Introduction **2**

2 Market data **2**
2.1 Instruments and conventions . . . 2
2.2 The production book . . . 2
2.3 Data pipeline . . . 3

3 The models **3**
3.1 Composition . . . 3
3.2 Variance blocks . . . 3
3.3 Jump laws . . . 4

4 Fourier pricing **4**
4.1 The COS method . . . 4
4.2 The PROJ method . . . 5
4.3 Validation of the pricers . . . 5

5 The calibration problem **5**
5.1 Loss functions . . . 5
5.2 Search space . . . 6
5.3 Optimisers . . . 6
5.4 Several days at once . . . 6

6 Analysis tools **6**
6.1 Model Greeks . . . 6
6.2 Volatility surface . . . 6
6.3 Gamma exposure . . . 7
6.4 Exotic options by Monte Carlo . . . 7

7 Results **7**
7.1 A calibrated day . . . 7
7.2 Calibration history . . . 7

8 Engineering **8**

9 Limits **8**

1 Introduction

Bitcoin options trade mostly on Deribit, with expiries from one day to more than a year and a smile that differs from equity markets: the skew changes sign over time, the wings are steep at short maturities, and jumps of several percent are routine. A model calibrated to this surface must combine a stochastic variance (term structure, smile dynamics) and jumps (short-dated wings); which combination works best is an empirical question, and answering it requires calibrating many models on many days.

The application makes that experiment interactive and portable. Its design goals are:

- *no installation*: a static page, the pricers in JavaScript, every day of data in a file loaded on demand;
- *transparency*: each model shows its dynamics, its constraints and its loss; every numerical choice is adjustable;
- *speed*: a full evaluation of the loss on a production day takes about 1.3 ms, so sliders reprice the whole chain live and an optimiser runs about 650 evaluations per second;
- *reproducibility*: the default settings reproduce the production calibration book exactly.

2 Market data

2.1 Instruments and conventions

Deribit options are European, written on the BTC/USD index X and quoted in BTC (inverse options): the premium of a call is $C^{\mathrm{BTC}} = C^{\mathrm{USD}}/X$. For each expiry T a future (or, for daily expiries, an implied forward) gives the forward F_T. Prices are converted to dollars, $P = P^{\mathrm{BTC}} X$, and the implied rate of the maturity is $r = \ln(F_T/X)/T$. Options are then priced with Black-76 on the forward, $P = e^{-rT}\,\mathrm{B76}(F, K, T, \sigma)$, and implied volatilities are obtained by inverting Black-76 (safeguarded Newton, bracket $[10^{-6}, 5]$, tolerance 10^{-10} on the price). Time to expiry is counted to the hour up to the 08:00 UTC expiry, which matters for the shortest maturities.

2.2 The production book

The target price of an option is its mark if bid $\le$ mark $\le$ ask, otherwise the mid. The production filters keep out-of-the-money options (puts with $K < F$, calls with $K > F$), quoted on both sides, with a relative spread (ask $-$ bid)/mid $< 20\,\%$ and a target price above 0.005 BTC, on the last six maturities beyond ten days. On 26 August 2026 this leaves 222 of the 914 options of the chain. Every filter can be changed in the application, including maturity-dependent thresholds (a 0.004 BTC option is a normal wing price at one week, an illiquid tail at one year).

2.3 Data pipeline

A day is a JavaScript file `jours/<date>.js` holding the whole chain in columns (strikes, maturities, F, r, bid, ask, mark, their implied volatilities, open interest) and the parameter sets calibrated that day. The page loads it with a `<script>` element, so it also works from the local disk. A GitHub Action downloads the chain from the public Deribit API every morning after the 08:00 UTC expiry, adds the day and rebuilds the page; the "Today" button does the same from the browser. The data cover 270 days since January 2024 and 529 calibrated parameter sets.

3 The models

3.1 Composition

Under the risk-neutral measure the future is a martingale. All models write

$$\ln \frac{F_T}{F_0} = X_T^c + X_T^J - \omega T, \tag{1}$$

where X^c is the continuous part driven by the variance block, X^J a pure-jump Lévy process independent of it, and ω the compensator that makes F a martingale.

Proposition 1. *Let ψ_J be the Lévy exponent of X^J, $\mathbb{E}[e^{\mathrm{i}uX_T^J}] = e^{T\psi_J(u)}$, and $A_{\mathrm{var}}(u,T)$ the log characteristic function of X_T^c, with $\mathbb{E}[e^{X_T^c}] = 1$. Then*

$$\varphi_T(u) = \mathbb{E}\big[e^{\mathrm{i}u\ln(F_T/F_0)}\big] = \exp\Big(A_{\mathrm{var}}(u,T) + T\psi_J(u) - \mathrm{i}u\,\omega T\Big), \qquad \omega = \psi_J(-\mathrm{i}), \tag{2}$$

and F is a martingale iff $\psi_J(-\mathrm{i})$ is finite.

Proof. Independence factorises the characteristic function. $\mathbb{E}[F_T/F_0] = \mathbb{E}[e^{X_T^c}]\,\mathbb{E}[e^{X_T^J}]\,e^{-\omega T} = e^{T\psi_J(-\mathrm{i})-\omega T}$, which equals one for all T iff $\omega = \psi_J(-\mathrm{i})$; the Markov property of (F, v) then gives the martingale property at all intermediate dates. □

The page assembles the pricer object from the two blocks: a block that is absent leaves no parameter at all ("no jumps" removes λ from the sliders, the bounds and the exports, it is not $\lambda = 0$).

3.2 Variance blocks

Black–Scholes. $A_{\mathrm{var}} = -\tfrac{1}{2}\sigma^2 T(u^2 + \mathrm{i}u)$.

Heston. $\mathrm{d}F_t/F_t = \sqrt{v_t}\,\mathrm{d}W_t$, $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \sigma\sqrt{v_t}\,\mathrm{d}Z_t$, $\mathrm{d}\langle W, Z\rangle_t = \rho\,\mathrm{d}t$. The exponent is affine in v_0, $A_{\mathrm{var}} = C(u,T) + D(u,T)v_0$, with the Riccati solution written in the form that avoids the branch-cut discontinuity of the complex logarithm (Albrecher et al. 2007):

$$d = \sqrt{(\rho\sigma\mathrm{i}u - \kappa)^2 + \sigma^2(u^2 + \mathrm{i}u)}, \qquad g = \frac{\kappa - \rho\sigma\mathrm{i}u - d}{\kappa - \rho\sigma\mathrm{i}u + d}, \tag{3}$$

$$C = \frac{\kappa\theta}{\sigma^2}\Big[(\kappa - \rho\sigma\mathrm{i}u - d)T - 2\ln\frac{1 - ge^{-dT}}{1 - g}\Big], \qquad D = \frac{\kappa - \rho\sigma\mathrm{i}u - d}{\sigma^2}\,\frac{1 - e^{-dT}}{1 - ge^{-dT}}. \tag{4}$$

The Feller condition $2\kappa\theta \geq \sigma^2$ keeps v strictly positive; the optimiser can enforce it.

Double Heston. Two independent CIR factors (Christoffersen, Heston and Jacobs 2009): the exponents add. A fast factor (large κ_1) shapes the short end, a slow one the long end; Feller is checked factor by factor.

Heston with variance jumps. $\mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \sigma\sqrt{v_t}\,\mathrm{d}Z_t + \mathrm{d}J^v_t$ with J^v compound Poisson of intensity λ_v and exponential sizes of mean μ_v (Duffie, Pan and Singleton 2000). The exponent gains

$$\lambda_v \int_0^T \Big(\frac{1}{1 - \mu_v D(u,t)} - 1 \Big)\,\mathrm{d}t, \tag{5}$$

computed by Simpson's rule on 64 intervals. The exponential law is what keeps the problem affine with a one-dimensional integral; a lognormal size would require a numerical integral inside the Fourier inversion, too slow for live use.

3.3 Jump laws

Table 1 lists the Lévy exponents. Their constraints are those of the existence of $\psi_J(-\mathrm{i})$, i.e. of $\mathbb{E}[e^{X^J_1}]$.

law	$\psi_J(u)$	constraint
Merton	$\lambda\big(e^{\mathrm{i}u\mu_J - \sigma_J^2 u^2/2} - 1\big)$	$\lambda \ge 0$, $\sigma_J > 0$
bi-lognormal	sum of two Merton exponents	$\lambda_1, \lambda_2 \ge 0$
Kou	$\lambda\Big(\frac{p\eta_1}{\eta_1 - \mathrm{i}u} + \frac{(1-p)\eta_2}{\eta_2 + \mathrm{i}u} - 1\Big)$	$\eta_1 > 1$, $\eta_2 > 0$
CGMY	$C\,\Gamma(-Y)\big[(M - \mathrm{i}u)^Y - M^Y + (G + \mathrm{i}u)^Y - G^Y\big]$	$M > 1$, $G > 0$, $Y < 2$, $Y \notin \{0, 1\}$
Variance Gamma	$-\frac{1}{\nu}\ln\Big(1 - \mathrm{i}u\theta\nu + \frac{1}{2}\sigma^2\nu u^2\Big)$	$1 - \theta\nu - \frac{1}{2}\sigma^2\nu > 0$
NIG	$\delta\Big(\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + \mathrm{i}u)^2}\Big)$	$\alpha > \lvert\beta + 1\rvert$, $\delta > 0$

Table 1: Lévy exponents of the jump blocks (Merton 1976; Kou 2002; Carr, Geman, Madan and Yor 2002; Madan, Carr and Chang 1998; Barndorff-Nielsen 1997).

Remark 1 (Identification). Several compositions have flat directions that no optimiser can resolve from one day of data: the two components of the bi-lognormal law are interchangeable unless their intensities live in disjoint ranges (production bounds $\lambda_1 \in [1, 300]$, $\lambda_2 \in [0, 1]$); in Heston, v_0 and θ trade off at short horizons and κ and σ along a ridge; jump intensity and jump size trade off for the short-dated wings. The page makes these visible: parameters can be locked, swapped ("Swap components 1 and 2") or freed per day when several days are calibrated together (section 5.4).

4 Fourier pricing

Both methods need only φ_T; the first two cumulants c_1, c_2 of $\ln(F_T/F_0)$ are obtained by central differences of $\ln\varphi_T$ at $u = 0$ (step 10^{-5}), and the fourth c_4 in closed form for each jump family.

4.1 The COS method

Fang and Oosterlee (2008) expand the density of $x = \ln(F_T/F_0)$ on a truncated interval $[a, b]$ in a cosine series whose coefficients are read directly from the characteristic function:

$$f(x) \approx \frac{2}{b-a}\sum_{k=0}^{N-1}{}' \,\mathrm{Re}\Big[\varphi_T(u_k)e^{-\mathrm{i}u_k a}\Big]\cos\big(u_k(x-a)\big), \qquad u_k = \frac{k\pi}{b-a}, \tag{6}$$

where $\sum'$ halves the first term. Integrating a payoff against this expansion uses the closed-form cosine moments χ_k, ψ_k of e^x and 1 on sub-intervals. Puts are priced (their payoff is bounded) and calls follow from put–call parity $C - P = e^{-rT}(F - K)$, which removes the cancellation error of deep in-the-money calls. The implementation uses $[a, b] = c_1 \pm L\sqrt{c_2}$ with $L = 24$ and $N = 512$

by default. Since all options of a maturity share $\varphi_T(u_k)$, these N values are computed once per maturity; the series is cut where a bound on its remainder falls below 10^{-10} \$: long maturities, whose characteristic function decays fast, need only a fraction of the terms.

Errors. The error has two parts: truncation of the density to $[a, b]$, controlled by L and the tails, and the series truncation, which decays exponentially for densities that are smooth on $[a, b]$. COS needs a valid second cumulant and loses accuracy when the jump part dominates the variance at very short maturities, where the density is spiky.

4.2 The PROJ method

Kirkby (2015) projects the density on a basis of linear B-splines on a uniform grid of N points and spacing Δ. The projection coefficients are obtained by one FFT of φ_T multiplied by the Fourier transform of the basis, and the payoff is integrated against the basis in closed form (with a three-point Gauss rule for the boundary cell). The grid half-width is

$$\alpha = \max\Big(L\sqrt{|c_2| + \sqrt{|c_4|}},\ 1.15 \max_K |\ln(K/F)| + c_1\Big), \tag{7}$$

$L = 20$, $N = 2^{11}$ by default. The error is $O(\Delta^2)$ for smooth densities and the method stays robust when $|\mu_J|$, σ_J or λ are large, which is why it serves as the reference for exotic controls and the volatility surface. COS is about three times faster for the calibration loop.

4.3 Validation of the pricers

- Put–call parity at the money: $C(K = F) - P(K = F) = 0.0000$ bp on the 28 compositions (martingale property).
- Black–Scholes recovered to within 0.005 bp; COS and PROJ equal on a full smile.
- JavaScript pricers equal to the Python reference pricers to $6 \cdot 10^{-6}$ \$.
- Successive optimisations of the code checked against the previous version on all compositions, COS and PROJ: equal to rounding noise ($4.5 \cdot 10^{-8}$ relative).

5 The calibration problem

5.1 Loss functions

For options i with target price P_i, implied volatility σ_i and Black-76 vega $\mathcal{V}_i$ (at the target volatility), and model values $\hat{P}_i(\Theta)$, $\hat{\sigma}_i(\Theta)$, the loss is a weighted mean of squared errors

$$\mathcal{L}(\Theta) = \frac{\sum_i w_i\, e_i(\Theta)^2}{\sum_i w_i}, \tag{8}$$

with five error measures (price, relative price, price over the bid–ask range, implied volatility in bp, and a combination of price and volatility with a weight k) and five weightings (none, raw $1/\mathcal{V}_i^2$, $1/\mathcal{V}_i^2$ normalised per maturity to mean one or to sum one, equal weight per maturity). The page displays the formula being minimised.

Proposition 2 (Vega weighting)**.** *To first order,* $P_i - \hat{P}_i \approx \mathcal{V}_i\,(\sigma_i - \hat{\sigma}_i)$*, so* $\sum_i (P_i - \hat{P}_i)^2/\mathcal{V}_i^2 \approx \sum_i (\sigma_i - \hat{\sigma}_i)^2$.

The price loss with $1/\mathcal{V}^2$ weights therefore approximates the implied-volatility loss without inverting Black-76 at each evaluation. The production loss, `inv_vega2_equal`, normalises the weights to sum one per maturity so that every maturity counts equally whatever its number of options; with raw weights, short maturities (small vega) dominate. A floor of 10^{-4} on the vega avoids infinite weights. The IV and price RMSE are reported whatever the loss.

5.2 Search space

Each free parameter is mapped to a slider $t \in [0,1]$ inside its bounds $[\ell, h]$: $\Theta = \ell(h/\ell)^t$ for positive scale parameters (log scale), $\Theta = \ell + t(h-\ell)$ for ρ and μ_J, and $\Theta = \ell + 0.01\,(e^{t\ln(1+(h-\ell)/0.01)} - 1)$ for intensities, which can reach zero exactly. The optimisers work in t; leaving $[0,1]$, or violating Feller when it is enforced, gives $\mathcal{L} = +\infty$. The default bounds are very wide; the production bounds of each model can be loaded.

5.3 Optimisers

- *Classic Nelder–Mead* (reflection, expansion, contraction, shrink coefficients $1, 2, \frac{1}{2}, \frac{1}{2}$), from the current parameters, initial step 0.04 in slider units, stopping when $(f_{\max} - f_{\min})/f_{\min} \leq 10^{-6}$ or after 2500 evaluations.
- *Adaptive Nelder–Mead* (Gao and Han 2012): coefficients 1, $1+2/n$, $\frac{3}{4} - \frac{1}{2n}$, $1 - \frac{1}{n}$ in dimension n, which keeps the simplex from degenerating when many parameters are free.
- *Differential evolution* (Storn and Price 1997): a population of popsize $\times\, n$ points initialised by Latin hypercube, mutation $v = x_{r_1} + F(x_{r_2} - x_{r_3})$ (rand1bin) or $v = x_{\text{best}} + F(x_{r_1} - x_{r_2})$ (best1bin) with F drawn in $[F_{\min}, F_{\max}]$ (dithering), binomial crossover with rate CR and greedy selection; it stops when the standard deviation of the population's losses falls below $\text{atol} + \text{tol}\,|\text{mean}|$. The production settings are rand1bin, $F \in [0.7, 1.5]$, $CR = 0.6$, popsize 25, tol 10^{-7}; the current parameters can be injected in the population.
- *DE then Nelder–Mead*: the Python production chain (global then local), with Nelder–Mead in place of SLSQP.

Same seed, same settings, same data: same result. The optimisers run in 12 ms slices between two frames, so the page stays fluid (60 frames per second) during a calibration.

5.4 Several days at once

A contract observed on two consecutive days gives two maturities T and $T - 1/365$: stacking N days multiplies the points of the term structure by N without new data. Parameters are shared by all days, which is the identifying assumption; the state variables (v_0, θ, ρ, jump intensities) can be freed day by day. The weighting can give each maturity, or each day, a total weight of one.

6 Analysis tools

6.1 Model Greeks

Greeks are computed by central finite differences on the model's own Fourier pricer, with the parameters fixed: Δ and Γ by moving F by $\pm 0.1\,\%$, vega by scaling the instantaneous volatility σ_0 by $\pm 0.1\,\%$ (σ in Black–Scholes, $\sqrt{v_0}$ in Heston, $\sqrt{v_{0,1} + v_{0,2}}$ in double Heston), theta by moving the maturity by one day on each side. With the Black–Scholes block, these differences recover the Black-76 closed forms to the decimals shown for Δ, Γ and vega, and within $0.1\,\%$ for Θ. The gap between the model delta and the Black-76 delta at the market volatility is the smile effect. With its parameters fixed, the model's implied volatility is a function of moneyness, so when F moves the option slides along the smile: $\Delta_{\text{model}} = \Delta_{\text{B76}} + \mathcal{V}\,\partial\sigma_{\text{imp}}/\partial F$. The sign of the correction follows the slope of the smile, and it is the extra hedge the model recommends.

6.2 Volatility surface

The model surface is priced by PROJ on a strike grid per maturity (25 to 61 strikes) and inverted with Black-76. It is shown either on the calibrated range only or over all the day's options, where it is an extrapolation of what the model has seen.

6.3 Gamma exposure

The gamma exposure of option i is $\mathrm{GEX}_i = s_i \cdot \mathrm{OI}_i \cdot \Gamma_i \cdot F^2 \cdot 1\,\%$, the dollar delta to trade to hedge a $+1\,\%$ move of the future, with open interest in BTC. The "market makers" convention takes $s_i = +1$ for calls and -1 for puts (dealers assumed long the public's calls and short its puts). The profile recomputes the total GEX for a future moved from $-20\,\%$ to $+20\,\%$ with the volatilities of each strike frozen; its zero is the *flip level*. It is a convention: Deribit does not publish who holds what.

6.4 Exotic options by Monte Carlo

Paths of the future are simulated with the active model. The CIR variance uses the quadratic–exponential scheme of Andersen (2008): given v_t, the conditional mean m and variance s^2 of $v_{t+\Delta}$ are matched by $a(b+Z)^2$ when $s^2/m^2 \leq 1.5$ and by a mixture of a mass at zero and an exponential law otherwise. Lognormal, Kou, Variance Gamma (gamma subordinator) and NIG (inverse Gaussian subordinator) jumps are simulated exactly per step and compensated, so the future stays a martingale; CGMY has no simple exact simulation and is excluded. Averages, extrema and barriers are monitored at each step. American options use Longstaff–Schwartz with the basis $1, x, x^2, x^3$, $x = F/K$, on in-the-money paths. The European with the same strike is simulated on the same paths and compared with its Fourier price: the gap measures the discretisation bias, and the European serves as a control variate, $\hat{Y}_{\mathrm{cv}} = \hat{Y} - b^\star(\hat{C} - C_{\mathrm{Fourier}})$ with $b^\star = \mathrm{Cov}(Y, C)/\,\mathrm{Var}(C)$. On six model families the simulated European falls within two standard errors of the Fourier price; the early-exercise premium is zero for $r = 0$ and positive for $r > 0$, as it must be.

7 Results

7.1 A calibrated day

Table 2 and fig. 1 show the Bates model (Heston + Merton) on 26 August 2026, with its production calibration (differential evolution then local search, COS, price loss with `inv_vega2_equal` weights).

	κ	θ	σ	ρ	v_0	λ	μ_J	σ_J
value	46.03	0.1334	2.491	0.139	0.1630	0.149	−0.460	0.783

Table 2: Bates parameters on 2026-08-26: IV RMSE 37.3 bp and price RMSE 44.0 $ on 222 options, Feller margin $2\kappa\theta - \sigma^2 = 6.08$.

The fit has a clear reading. The variance reverts very fast ($\kappa = 46$, a half-life of $\ln 2/\kappa \approx 5.5$ days) towards $\sqrt{\theta} = 36.5\,\%$ from $\sqrt{v_0} = 40.4\,\%$, with a large volatility of variance: the term structure of the smile is short and steep. The spot–volatility correlation is *positive* ($\rho = 0.14$), the opposite of equity indices: on that day upward moves were expected to come with higher volatility, which lifts the call wing. The jump component is a rare crash ($\lambda = 0.15$ per year, mean size $e^{-0.46} - 1 \approx -37\,\%$, dispersion 0.78), which carries the put wing at long maturities.

7.2 Calibration history

Figure 2 shows the best IV RMSE of each day among the calibrated sets embedded in the application, for Bates and for the bi-lognormal Bates model. The median is 35.9 bp for Bates (61 days) and 37.7 bp for the bi-lognormal model (73 days): the second jump component rarely pays for its three extra parameters on these days, but it removes the occasional failures of a single jump size (the worst Bates day is above 1000 bp, a failed local search, against 114 bp for the bi-lognormal model). The median correlation is slightly positive for both models.

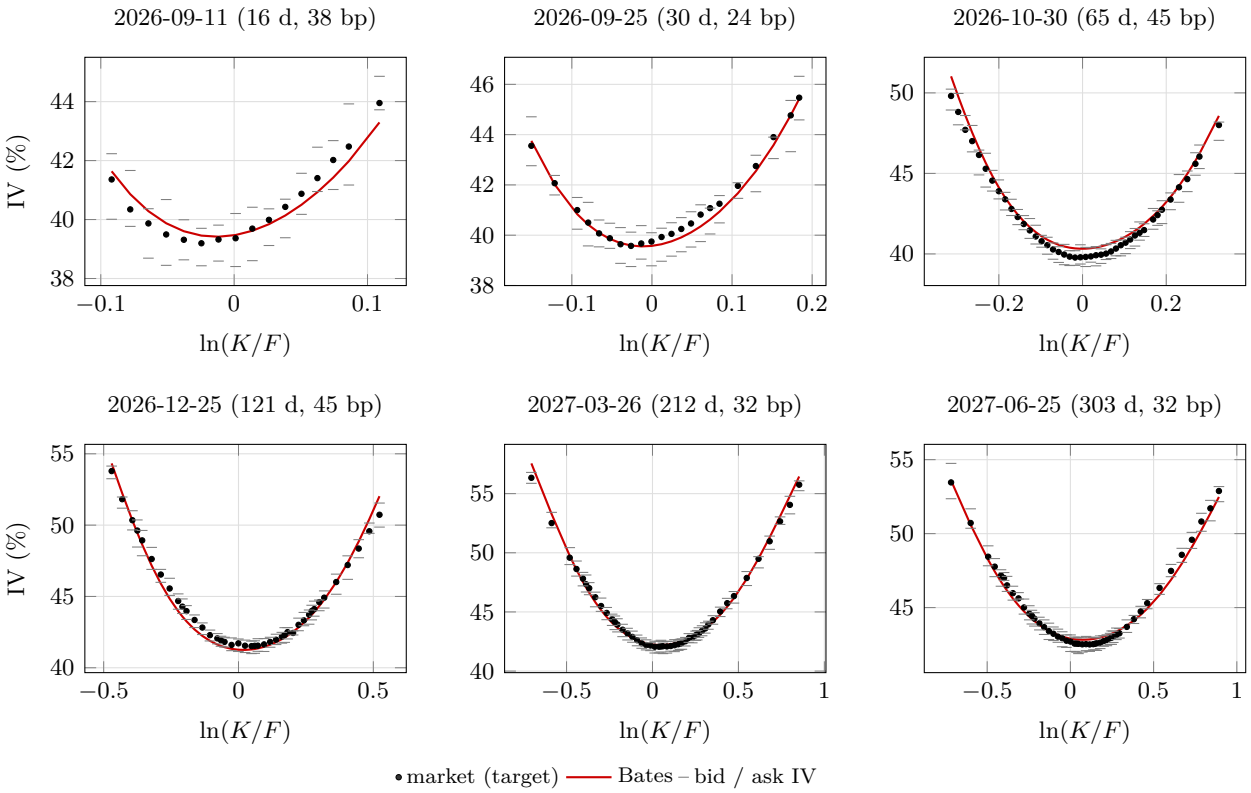


Figure 1: The six production maturities of 2026-08-26 and the calibrated Bates smiles; the IV RMSE of each maturity is in its title. Data exported from the application.

8 Engineering

- *A static page*: `index.html` carries the page, its pricers and one day; the other days, the futures history and the calibrated sets are files loaded on demand. Settings are kept in the browser, or written to a folder of the disk with the File System Access API (Chrome, Edge). No server, no database, no account.
- *Performance* (Chrome, Bates, six maturities, 222 options, COS): a full evaluation of the loss takes about 1.3 ms, a screen refresh about 3.5 ms. Characteristic-function constants that do not depend on u are computed once per parameter set; complex arithmetic is written in real numbers in the hot loops; cosines are advanced by rotation instead of being recomputed.
- *Mathematics on the page*: formulas are written in LaTeX and rendered once at build time with KaTeX, so the page runs no math script; rendered formulas are cached, and the daily GitHub Action needs only Python.

9 Limits

- The local optimisers depend on their starting point; differential evolution is global but far more expensive.
- Stacking days assumes parameters stable over the window; sharing v_0 is justified only for close days.
- Several compositions are weakly identified on a single day (section 3); a good RMSE does not mean that every parameter is meaningful.
- Writing files works only in browsers that implement the File System Access API.

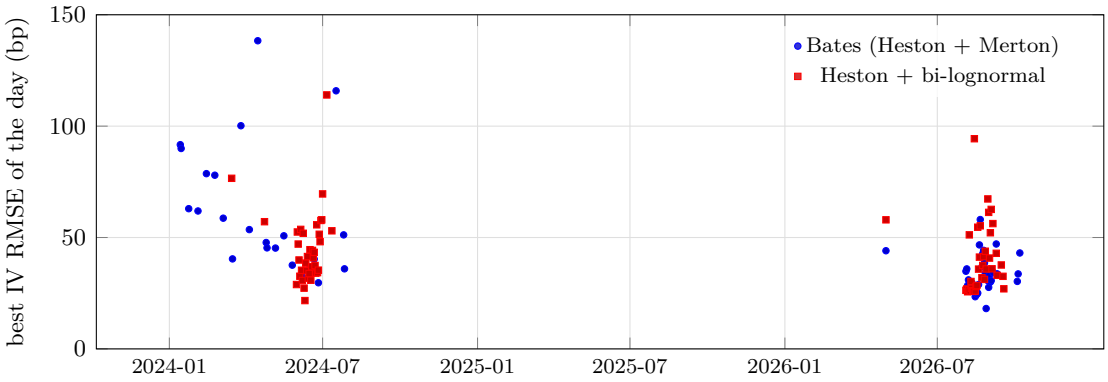


Figure 2: Best IV RMSE per day of the calibrated sets embedded in the application (values above 150 bp not shown). The gap in 2025 is a gap in the calibration campaigns, not in the market data.

References

[1] H. Albrecher, P. Mayer, W. Schoutens and J. Tistaert. The little Heston trap. *Wilmott Magazine*, January 2007.

[2] L. Andersen. Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3):1–42, 2008.

[3] O. E. Barndorff-Nielsen. Normal inverse Gaussian distributions and stochastic volatility modelling. *Scandinavian Journal of Statistics*, 24(1):1–13, 1997.

[4] D. Bates. Jumps and stochastic volatility: exchange rate processes implicit in Deutsche mark options. *Review of Financial Studies*, 9(1):69–107, 1996.

[5] P. Carr, H. Geman, D. Madan and M. Yor. The fine structure of asset returns: an empirical investigation. *Journal of Business*, 75(2):305–332, 2002.

[6] P. Christoffersen, S. Heston and K. Jacobs. The shape and term structure of the index option smirk: why multifactor stochastic volatility models work so well. *Management Science*, 55(12):1914–1932, 2009.

[7] D. Duffie, J. Pan and K. Singleton. Transform analysis and asset pricing for affine jump-diffusions. *Econometrica*, 68(6):1343–1376, 2000.

[8] F. Fang and C. Oosterlee. A novel pricing method for European options based on Fourier-cosine series expansions. *SIAM Journal on Scientific Computing*, 31(2):826–848, 2008.

[9] F. Gao and L. Han. Implementing the Nelder–Mead simplex algorithm with adaptive parameters. *Computational Optimization and Applications*, 51(1):259–277, 2012.

[10] S. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343, 1993.

[11] J. L. Kirkby. Efficient option pricing by frame duality with the fast Fourier transform. *SIAM Journal on Financial Mathematics*, 6(1):713–747, 2015.

[12] S. Kou. A jump-diffusion model for option pricing. *Management Science*, 48(8):1086–1101, 2002.

[13] F. Longstaff and E. Schwartz. Valuing American options by simulation: a simple least-squares approach. *Review of Financial Studies*, 14(1):113–147, 2001.

[14] D. Madan, P. Carr and E. Chang. The variance gamma process and option pricing. *European Finance Review*, 2:79–105, 1998.

[15] R. Merton. Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3:125–144, 1976.

[16] R. Storn and K. Price. Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, 11:341–359, 1997.